OPERATING SYSTEMS

Memory Management

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Paging

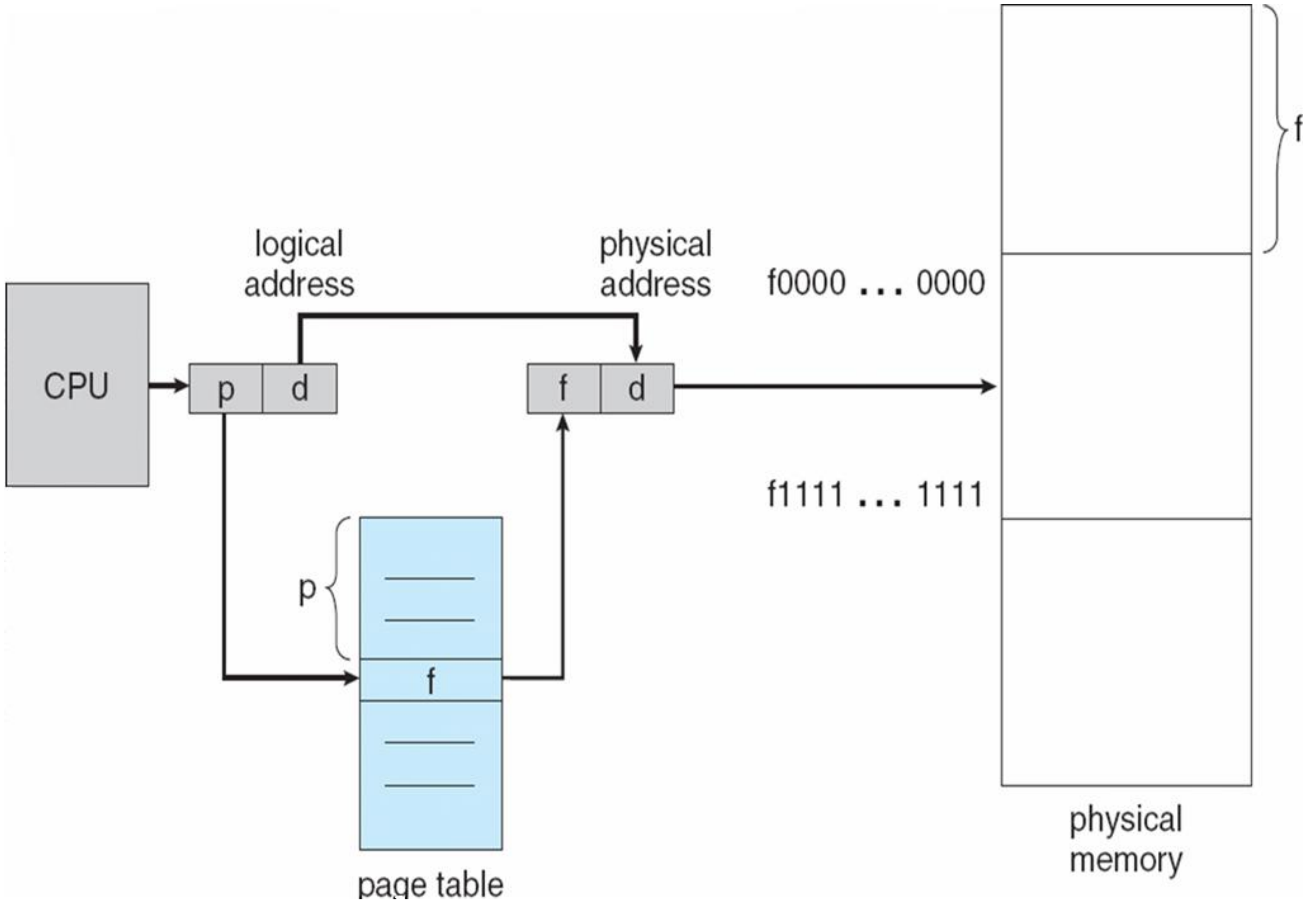
- Logical address space of a process can be noncontiguous; process is allocated physical memory whenever the latter is available
- **Divide physical memory into fixed-sized blocks called frames** (size is power of 2, between 512 bytes and 8,192 bytes)
- Divide logical memory into blocks of same size called pages
- Keep track of all free frames

Paging

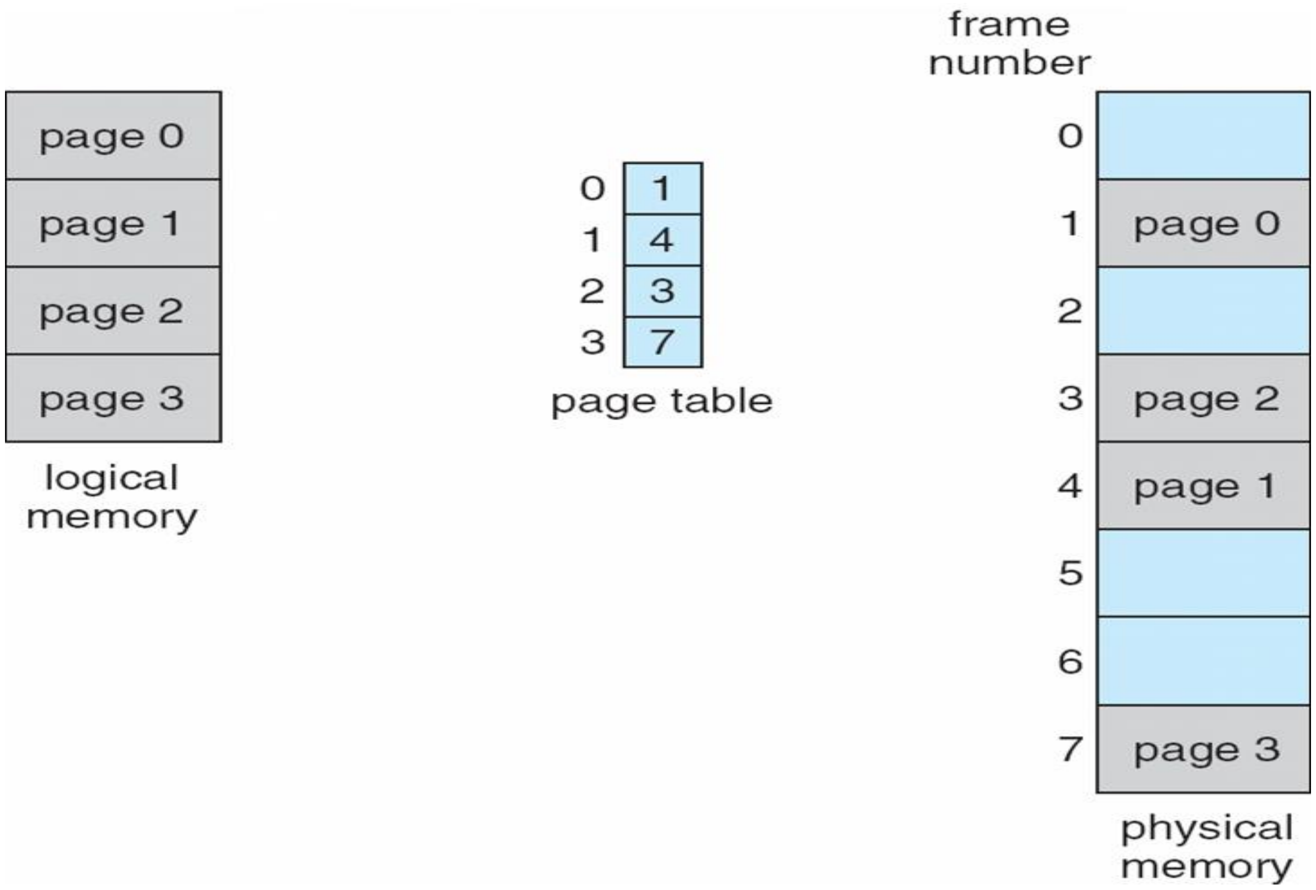
- To run a program of size n pages, need to find n free frames and load program
- Set up a page table to translate logical to physical addresses
- Internal fragmentation may occur in the last frame

- Address generated by CPU is divided into:
  - **Page number (*p*)** – used as an index into a *page table* which contains base address of each page in physical memory
  - **Page offset (d)** – combined with base address to define the physical memory address that is sent to the memory unit

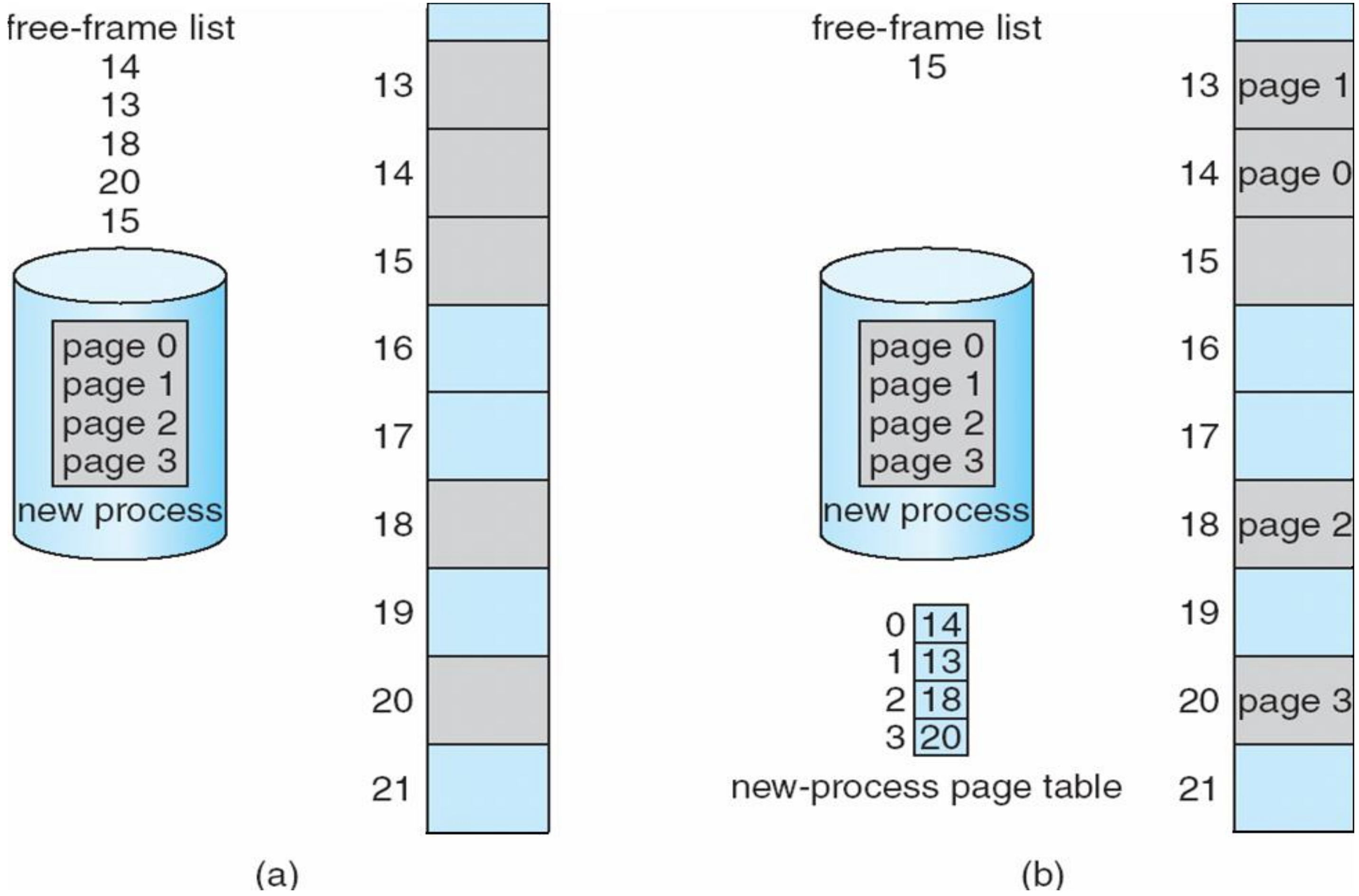
Paging Hardware



Physical Model of Logical and Physical Memory



Free Frames Before Allocation(a) and After Allocation(b)



Paging - Implementation

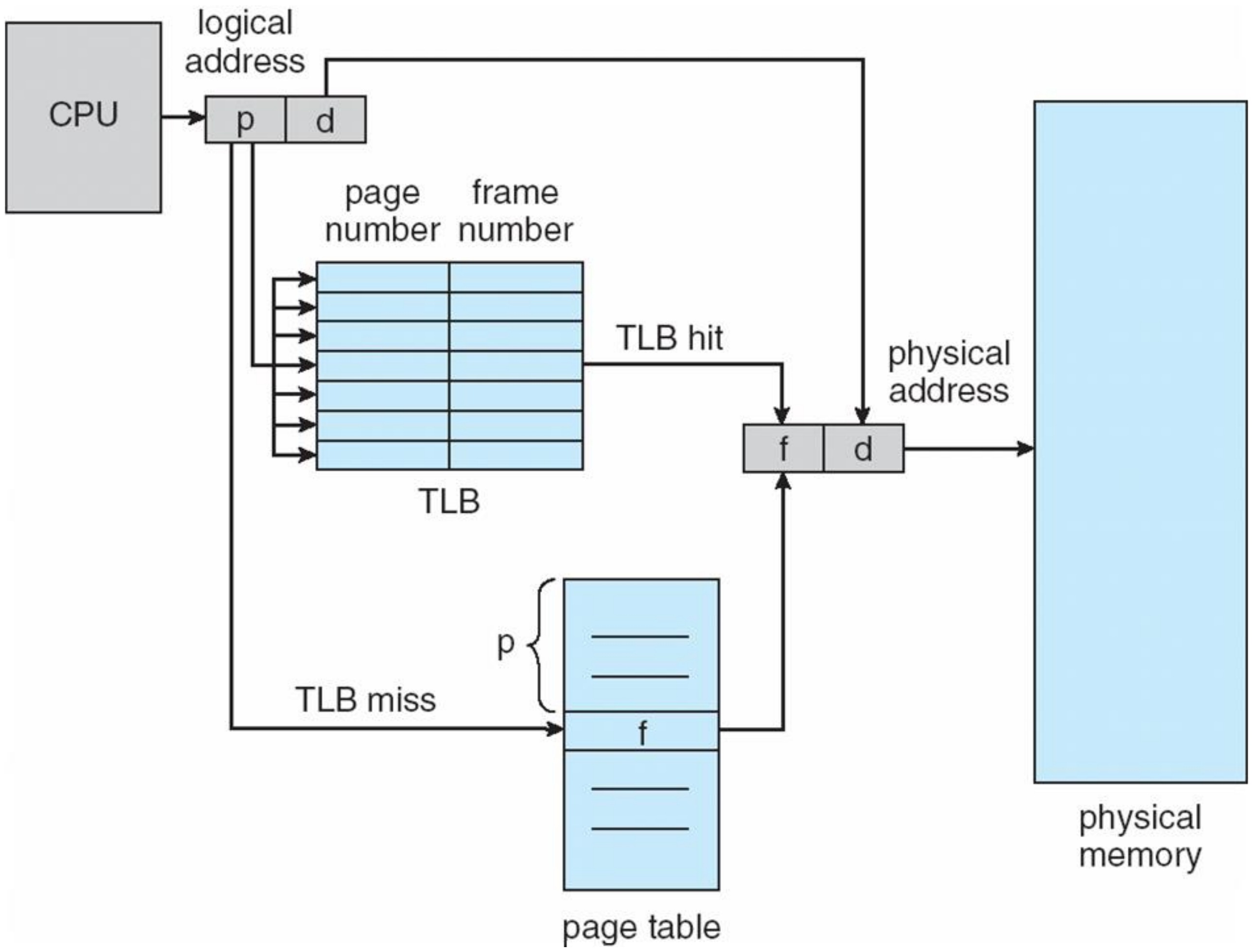
- Page table is kept in main memory
- Page-table base register (PTBR) points to the page table
- Page-table length register (PTLR) indicates size of the page table
- In this scheme, every data/instruction access requires two memory accesses. One for the page table and one for the data/instruction.

Paging - Implementation

- The two memory access problem can be solved by the use of a special fast-lookup hardware cache called associative memory or translation look-aside buffers (TLBs)
- Some TLBs store address-space identifiers (ASIDs) in each TLB entry – uniquely identifies each process to provide address-space protection for that process

OPERATING SYSTEMS

Paging - Implementation

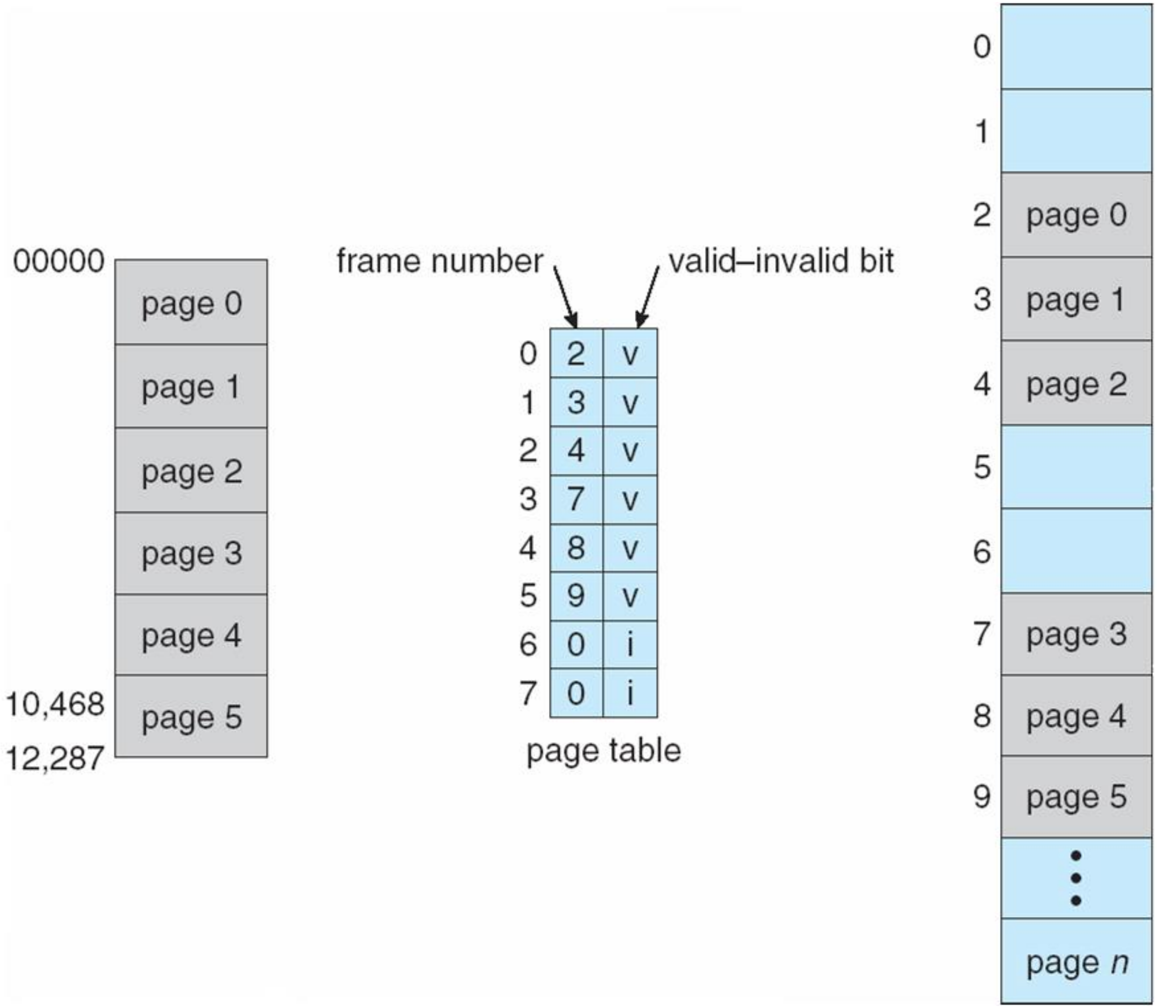


Memory Protection

- Memory protection implemented by associating protection bit with each frame

- **Valid-invalid** bit attached to each entry in the page table:
  - “valid” indicates that the associated page is in the process’ logical address space, and is thus a legal page
  - “invalid” indicates that the page is not in the process’ logical address space

Paging - Implementation



THANK YOU

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